

ITT320 INTRO TO COMPUTER SECURITY

Chapter 4 : Intrusion-Detection Systems (IDS)

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Objectives

- Explain how intrusion-detection systems work
- Implement strategies for preventing intrusion
- Identify and describe several popular intrusion-detection systems
- Define the term *honey pot*
- Identify and describe at least one honey pot implementation



Introduction to Intrusion Detection Systems (IDS)

An intrusion detection system (IDS) is a system that monitors network traffic for suspicious activity and alerts when such activity is discovered.

Intrusion-Detection Systems (IDS) allow system administrators to detect possible attacks to the network.

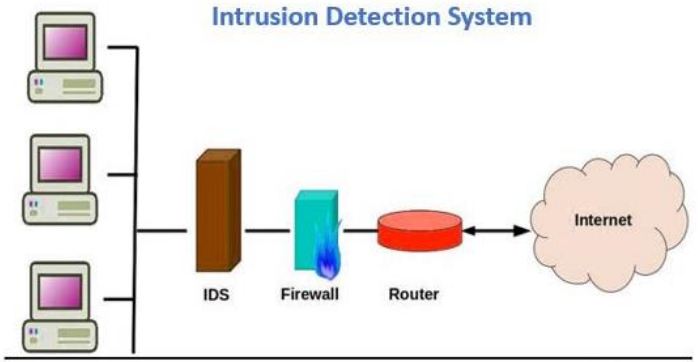
This chapter explores implementations of IDS solutions. We also explore the concept of a “honey pot” and how it can help administrators track attackers of the network.

What is IDS?

Intrusion detection systems are used to detect anomalies with the aim of catching hackers before they do real damage to a network.

Characteristics / How it Works:

- Network Host IDS – IDS Resides on the network
- Host Based IDS – IDS is installed on the client computers
- Capable of taking actions when malicious activities is detected
  - Blocking traffics sent from attacker IPs
- Please make your own research the different between IDS vs IPS

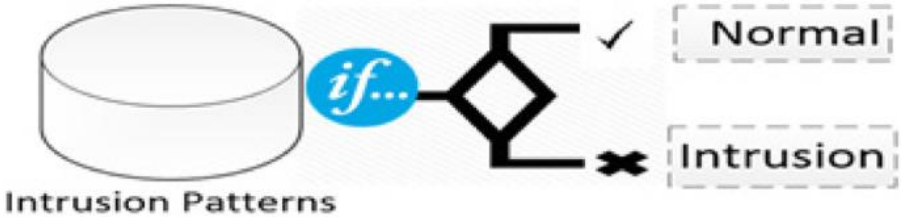


Source : forum.Huawei.com

Understanding IDS Concept

There are few IDS concepts that we can implement based on the situation, objectives, budget, expertise and goals.

1. Preemptive Blocking
2. Infiltration
3. Intrusion Deflection
4. Intrusion Deterrence
5. Anomaly Detection



Understanding IDS Concept
Pre-emptive Blocking

Pre-emptive Blocking – seeks to prevent intrusions **before** they occur.

Technique : detect the early foot printing stages, then **block** the IP or user of the source of foot printing activities.

- Sometimes called “banishment vigilance”
- Attempts to detect impending intrusions through foot printing
- Susceptible to false positives
- May block legitimate traffic



Understanding IDS Concept Infiltration

Infiltration – proactive effort by the administration or security specialist to get information from various sources like online/offline hacking group.

- The concept of going undercover in the hacker world
- Information is gathered through the hacker community

to find out what vulnerabilities are being exploited

Drawbacks:

- Time consuming
- Administrators are not trained in detective work
- Dangerous



deciso.com

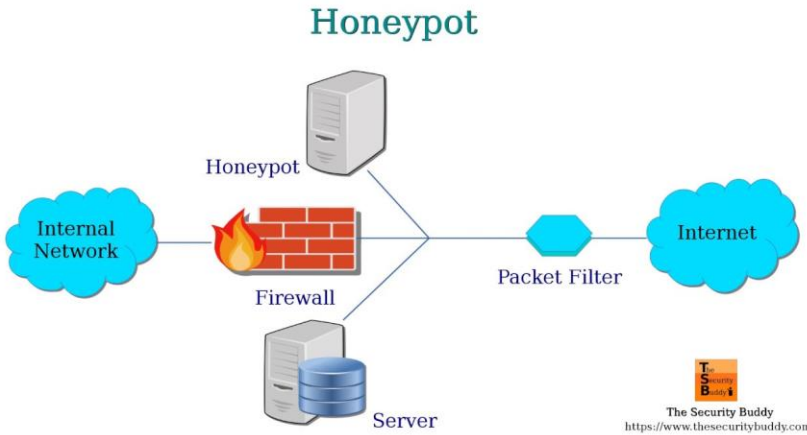
Understanding IDS Concept
Intrusion Deflection

Intrusion Deflection : the administrator create a fake attractive system / server environment for the purpose to attract hacker to get in. Monitor, get the proof and block the attacker.

- An attempted intrusion is redirected to a special environment and monitored
- Honeypots are used in this approach – set up a

Fake system/server that appears to be an entire subnet

- Difficult to setup and maintain
- Assumes a target system will be compromised



Understanding IDS Concept
Intrusion Deterrence

Intrusion Deterrence : try to make the system seem like a palatable target or difficult to attack, less attractive system, warning of active monitoring

- Making the reward of intrusion not worth the effort
- Incorporates hiding the important servers through the use of camouflage
- Uses multiple warnings to scare off attackers
- Low cost solution
- Easy to setup



Understanding IDS Concept
Anomaly Detection

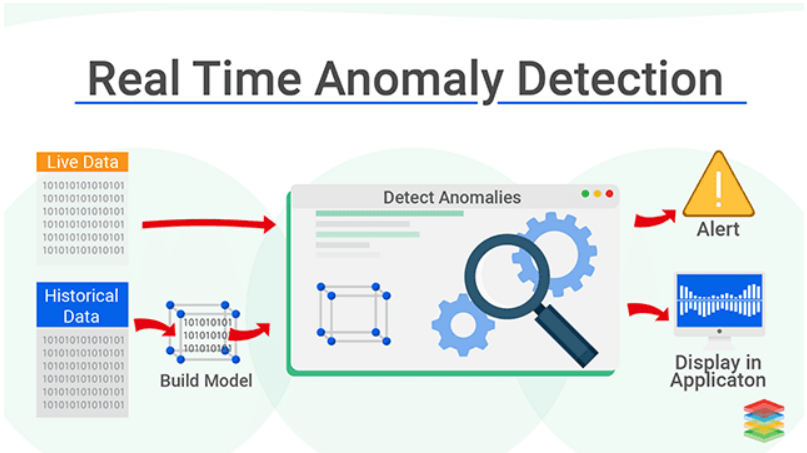
Anomaly Detection

Actual software works to detect intrusion attempts

And notify administrator. How? The system will looks for any anomalous behavior.

- Any activity that does not match normal use is saved in a log
- Normal usage profiles are kept and updated,

then compared to anomalous behavior



Understanding IDS Concept
Anomaly Detection

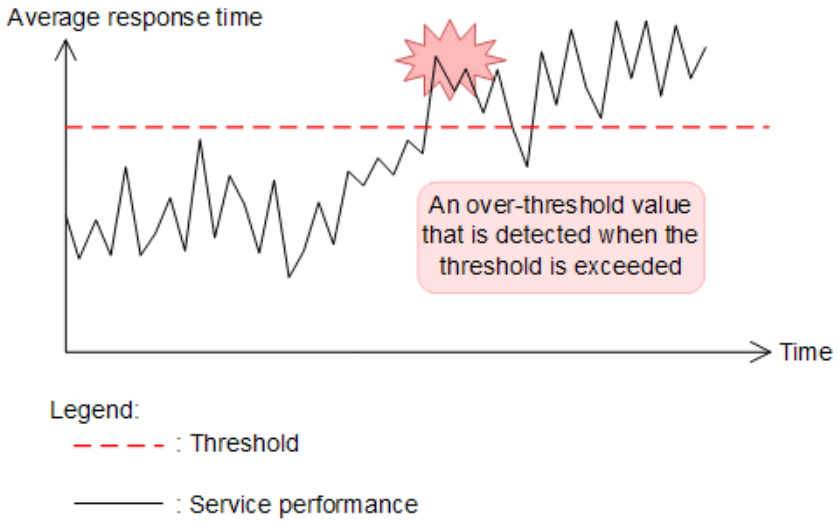
Anomalous Detection

Threshold Monitoring

- Define acceptable behaviors

(RAM/CPU/Bandwidth Usage in a Company)

- Presets acceptable behavior levels
- Monitors the exceeding of these levels
- Difficult to set times for monitoring behavior
- Susceptible to false positives and negatives

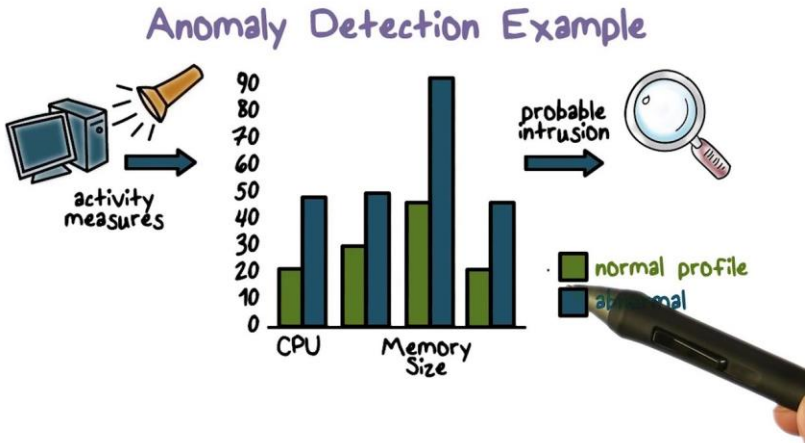


Understanding IDS Concept
Anomaly Detection

Anomalous Detection

Resource Profiling

- Develops historic usage profile, system-wide
- Difficult to interpret the meaning of changes in usage

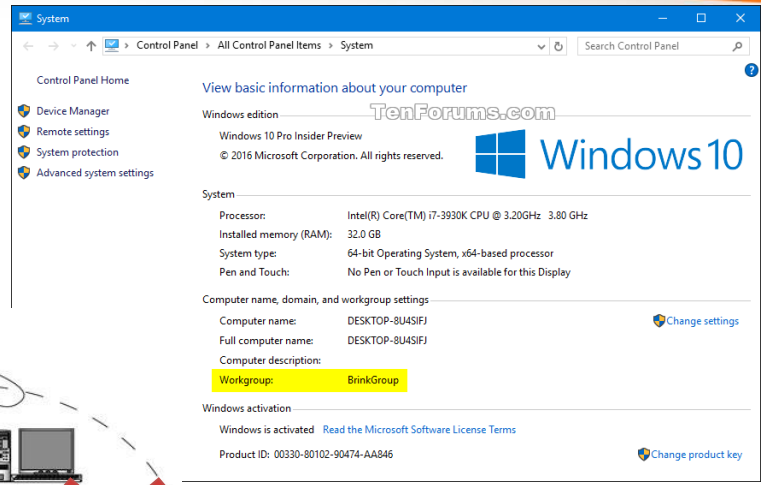


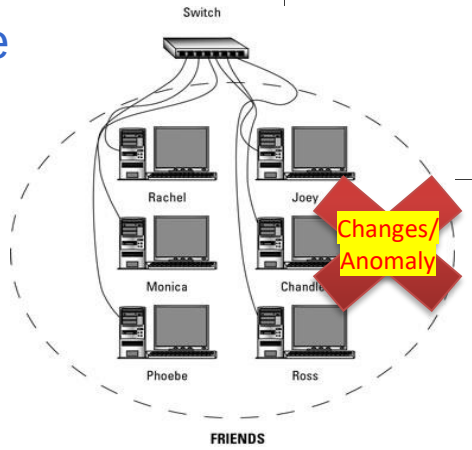
Understanding IDS Concept
Anomaly Detection

Anomalous Detection

User / Group Work Profiling

- Profiles are kept on a user or group level
- Changes in work patterns need to be updated in profile
- Dynamic user base could be difficult to profile



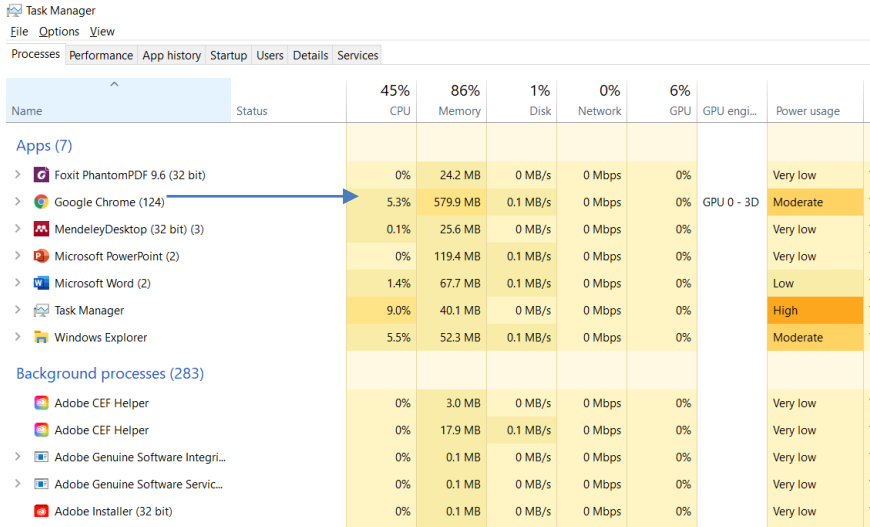


Understanding IDS Concept
Anomaly Detection

Anomalous Detection

Executable Profiling

- Monitors how programs use system resources
- System services cannot be traced to a particular user
- Profiles how system objects (files, printers) are normally used

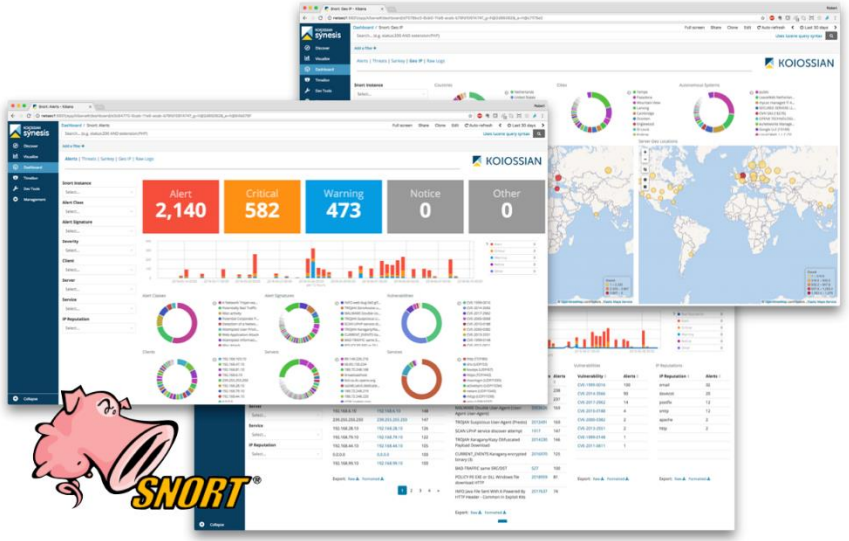


Understanding & Implementing IDS Systems

TWO Systems discussed in this sectior

- SNORT
- Cisco Intrusion-Detection





Understanding & Implementing IDS Systems - Snort

Snort is an open source network intrusion prevention and detection system, capable of performing real-time traffic analysis and packet logging on IP networks.

It is a software implementation installed on server to monitor incoming traffics.



https://snort.org

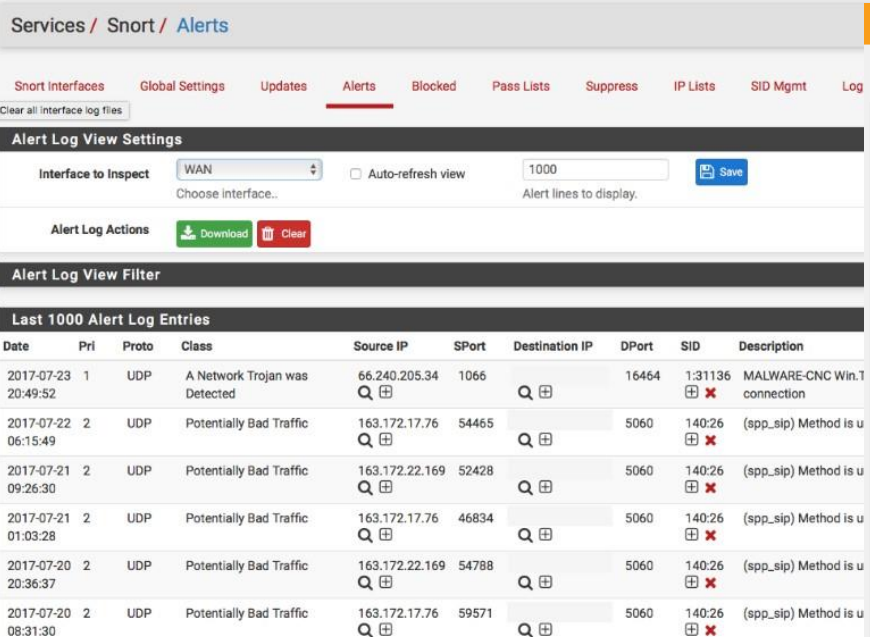
- Possibly the most well-known open source IDS
- Available on multiple platforms including:
  > UNIX, Linux, and Windows
- Three modes of operation:
  1. Sniffer
  2. Packet logger
  3. Network intrusion-detection

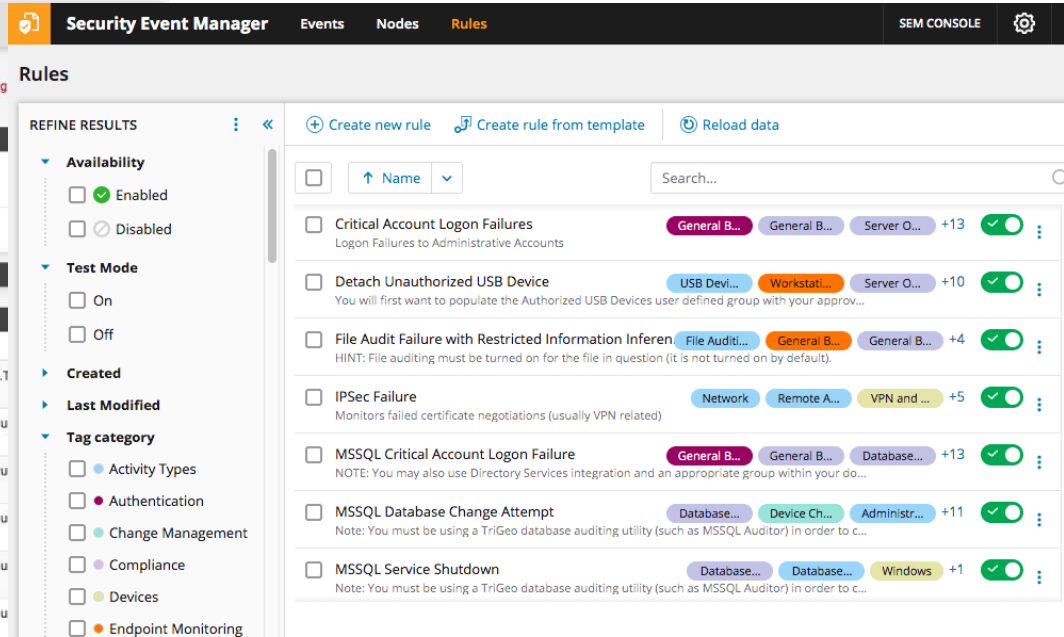
More at : https://www.dnsstuff.com/intrusion-detection-system

Understanding & Implementing IDS Systems - Snort

Snort Alert Logs View Interfaces

SolarWinds Snort Logs Analyzer



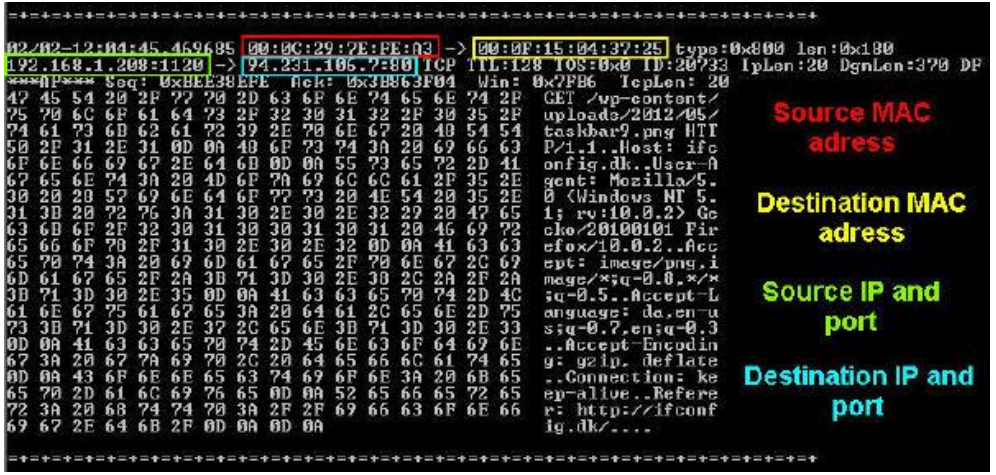


Understanding & Implementing IDS Systems - Snort

Sniffer Mode – the console will display a continuous stream of the contents of all packets coming across the machines

- Monitors all traffic coming and going on a computer
- Excellent way to check encryption of Data transmission
- Helps determine potential sources of problems

Source : ifconfig.dk



Understanding & Implementing IDS Systems - Snort

Packet Logger Mode

- Similar to sniffer mode
- Packet contents are written to a text file (Not in console / command line)
- Contents can be searched for specific items

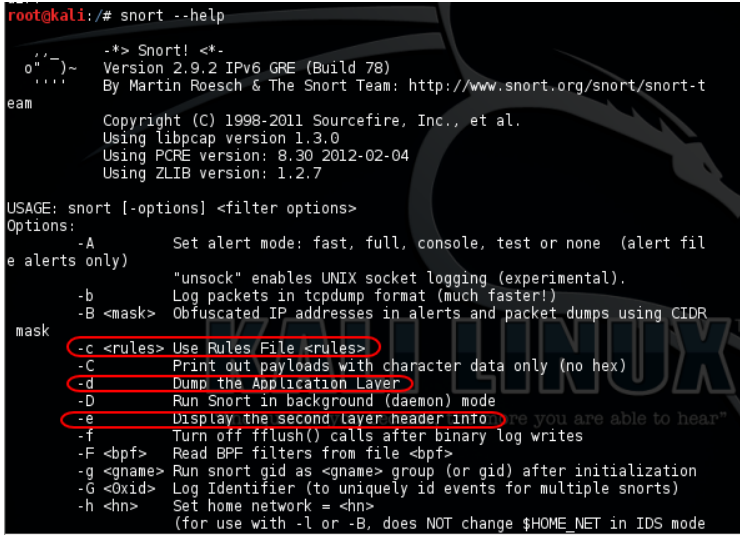


There are many security company that create the log analyzer tools Such as SolarWinds Security Event Manager.

Understanding & Implementing IDS Systems - Snort

Network Intrusion-Detection

- Uses a heuristic approach to detect anomalous traffics
- Rules-based – require configuration based on admin settings and requirements to achieve goals
- Command line based interface – must know the correct command and its output
- Need to know commands and what they do



Understanding & Implementing IDS Systems Cisco Intrusion-Detection

Cisco **intrusion detection systems** (**IDS**) and **intrusion prevention systems** (**IPS**) are some of many systems used as part of a defense-in-depth approach to protecting the network against malicious traffic.

Examples:

- ✓ Cisco IDS 4200 Series Sensors (Dedicated Appliances)
- ✓ Cisco Catalyst 6500 (Multilayer Switches)
- ✓ Series Intrusion Detection System Services Module (IDSM-2)

Updated:

- ✓ Cisco **FirePOWER 8000/7000** series appliances
- ✓ Virtual Next-Generation IPS (**NGIPSv**) for Vmware
- ✓ A **module** in an IOS router, such as the **AIM-IPS** or **NME-IPS** modules



Cisco IPS 4270 Sensor - *A sensor is a device that looks at traffic on the network and then makes a decision based on a set of rules to indicate whether that traffic is okay or whether it is malicious in some way*

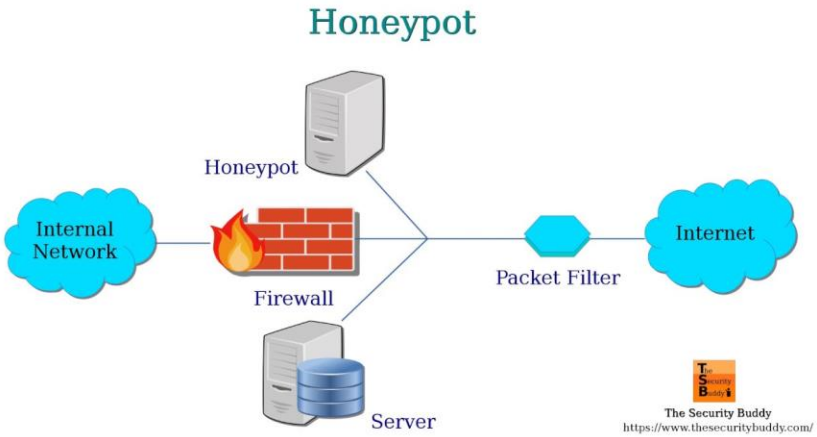
Understanding & Implementing IDS Systems – Honey Pots

What is?

A honeypot resides in a DMZ as a vulnerable host and advertises services and software to entice a hacker to hack the system.

- A honey pot is a single machine
- Set up to appear to be an important server
- All traffic to the machine is suspicious
- No legitimate users should connect
- Two types of Honey Pots discussed here:

1. **Specter**
2. Symantec Decoy Server





Understanding & Implementing IDS Systems – Specter

Specter is a honeypot system that can automatically capture information about a hacker's machine while they're attacking the system.

It simulates a vulnerable computer, providing an interesting target to trap hackers away from the production machines.

- Software solution, a smart honeypot-based intrusion detection system
- automatically investigates the attackers while they are still trying to break in
- provides massive amounts of decoy content including images, MP3 files, email messages, password files, documents and all kinds of software
- Able to emulate common services:
  - SMTP, FTP, TELNET, FINGER, POP3, etc
- Currently – there is no update / latest version of Specter and maybe will become private / paid tools by Deceptions Technologies



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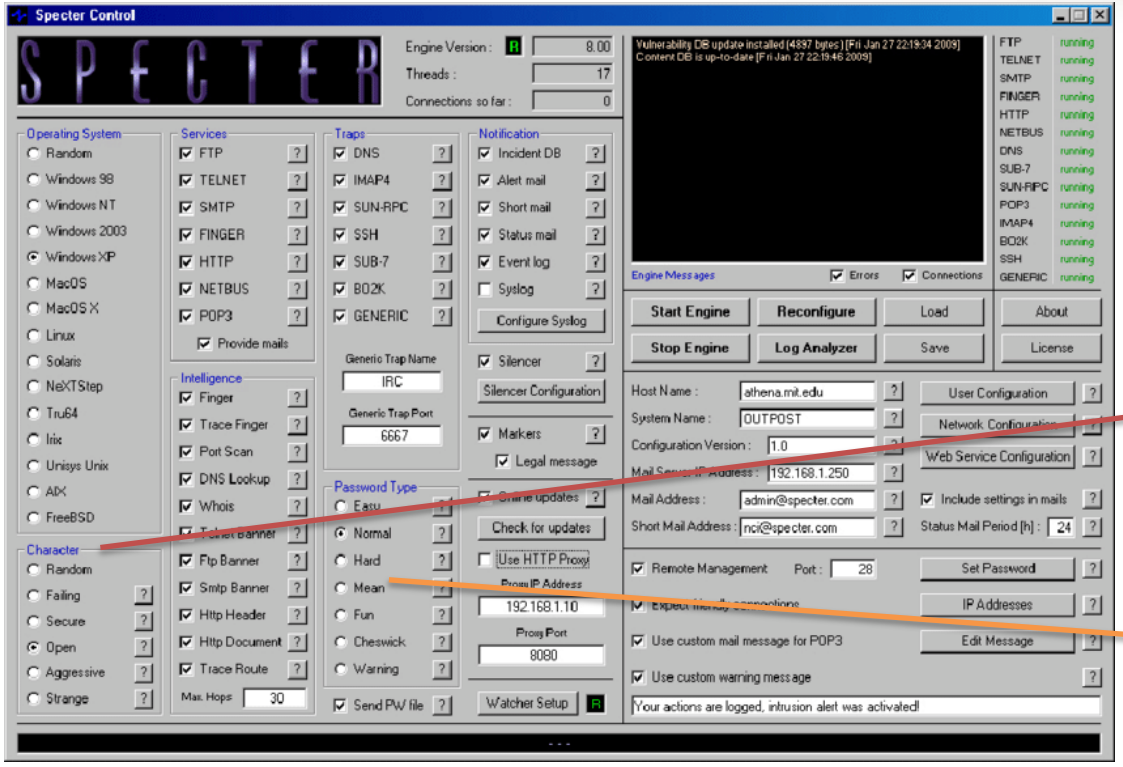


Understanding & Implementing IDS Systems – Specter

There are **5** different characters available for the simulated host:

1. **Open** - The system behaves like a badly configured system in terms of security
2. **Secure** - The system behaves like a well configured system in terms of security
3. **Failing** - The system behaves like a machine with various hard- and software problems
4. **Strange** - The system behaves unpredictable and leaves the intruder wondering what's going on
5. **Aggressive** - The system communicates as long as necessary to collect information about the attacker, then reveals its true identity by the appropriate means depending on the kind of connection and then ends communication. This is very handy to scare intruders away

Understanding & Implementing IDS Systems – Specter



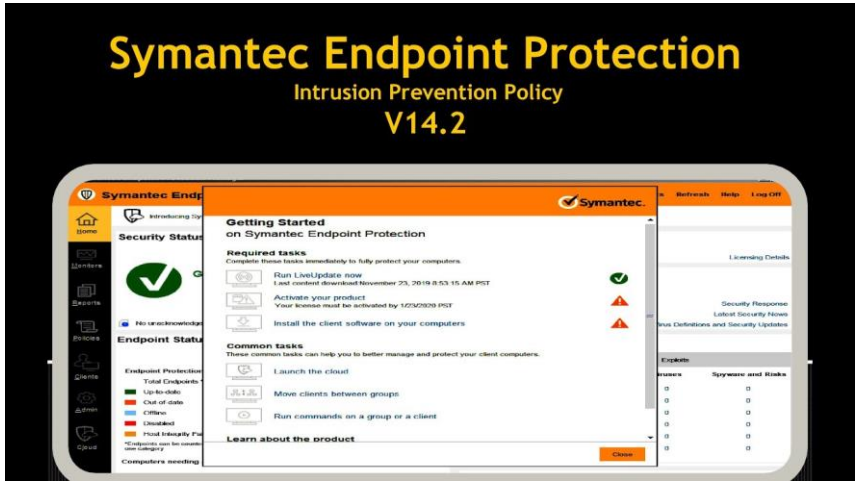
Can be set up in one of five modes:

1. Open
2. Secure
3. Failing
4. Strange
5. Aggressive

Fake Password Files Can Be Created:

EASY | HARD | NORMAL | FUN | WARNING

Understanding & Implementing IDS Systems Symantec Decoy Server



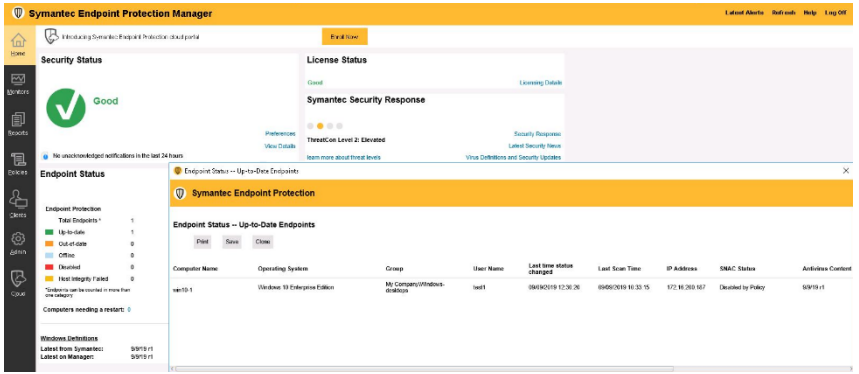
- Symantec Decoy Server provides early detection of threats and enables attack diversion and confinement by actually becoming the target of the attack
- The decoy sensor acts like a fully functioning server, and can simulate email traffic between users in the organization to mirror the appearance of a live mail server

Understanding & Implementing IDS Systems Symantec Decoy Server

When attacks are directed at the decoy sensor, Symantec Decoy Server delivers comprehensive attack detection through a system of data collection modules.

Full details can be found at:
http://enterprisesecurity.symantec.com/content/displaypdf.cfm?pdfid=292

Should be no surprise that Symantec provides a honey pot solution

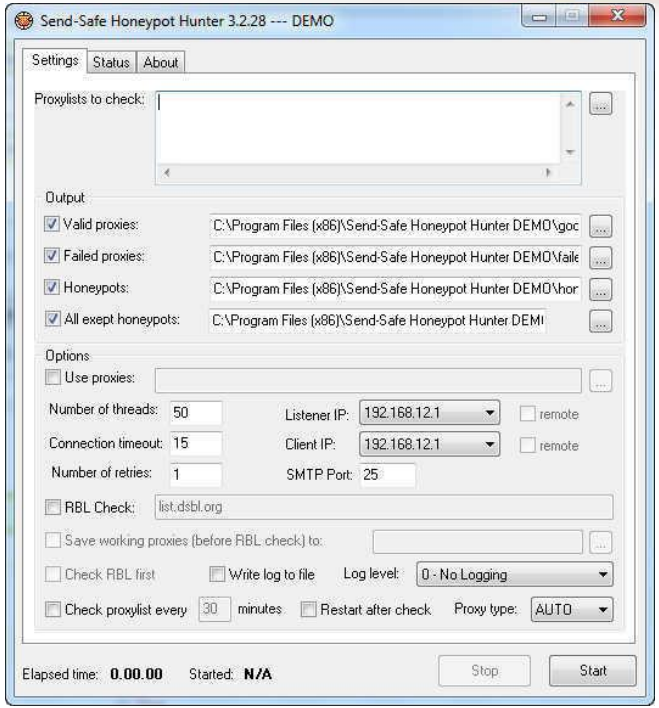


Oh No!
Honeypot Countermeasure

ALERT !

Hacker will avoid from being trapped into honeypots by using Anti-Honeypot software like Send-safe honeypot hunter

Nessus Vulnerability Scanner also can detect honeypot



Others for ODL

Other example of Honey Pots

- HoneyNets
- Nepenthes
- HoneyD (Virtual Honeypot)
- KFSensor
- BackOfficer Friendly

Please make your own study or choose as your project title



SUMMARY

- There are a variety of Intrusion Detection Systems available
- Should be used in conjunction with firewalls
- Can run at the perimeter and internally as sensors
- Ideally implemented on every server
- Free IDS solutions are available
- Honey Pots entice hackers to a fake server
- A server is set up specifically to monitor hacker activity
- Honey Pots can help track and catch hackers
- Honey Pots can be configured to emulate many server services

